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Conditional Cash Transfers as Stabilization Tool: a HANK Approach

1. Model

The setup is a standard New Keynesian model with incomplete markets and labor supply risk. Time is discrete and runs from $t = 0$ to infinity. There are four types of agents in the economy: households, firms, labor unions, and the government.

Households are heterogeneous (Aiyagari 1994), workers and firms join in a formal-informal setup, and the rest of the economy follows the textbook New Keynesian model (Galí 2015). There is no aggregate uncertainty and the economy starts in the deterministic steady-state.

Households

Assume there exists a continuum $i \in [0, 1]$ of ex-ante identical households, but experiencing idiosyncratic shocks in labor productivity and discount factor.

Households enjoy a CRRA-GHH utility over consumption $c_{i,t}$ and worked hours $h_{i,t}$ (Greenwood, Hercowitz, and Huffman 1988)¹, which we define $u(\tilde{c}_{i,t}) = u(c_{i,t} - v(h_{i,t}))$ as

$$\max_{c_{i,t}, h_{i,t}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{\tilde{c}_{i,t}^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} \right], \quad \text{with } v(h) = \psi \frac{h^{1+\frac{1}{\varphi}}}{1+\frac{1}{\varphi}}$$

where γ is the elasticity of intertemporal substitution (EIS) and φ is the Frisch elasticity of labor supply.

A household faces idiosyncratic labor productivity shocks $e_{i,t}$ with $\mathbb{E}[e_{i,t}] = 1$, and discount factor heterogeneity $\beta_i \in \{\beta_L, \beta_H\}$, and we fix ω_I impatient households. They can work in the formal or informal sector, or be unemployed $s \in \{F, I, U\}$. Each sector has a specific productivity $\theta_{i,t}^s$ which their wage depends on, re-drawn on each new offer, with $\mathbb{E}[\exp\{\theta_{i,t}^F\}] = 1.0$.

Formal workers earn labor income $(1 - \tau_\ell)\theta_{i,t}^F w_t e_{i,t} h_{i,t}$, while informal workers earn $\theta_{i,t}^I w_t e_{i,t} h_{i,t}$, where w_t is the baseline wage, τ_ℓ is a fixed labor income tax rate. Let $y_t(s, e, \theta)$

¹The GHH specification assumes zero income effect, so that the hours decision is separable from the consumption-savings problem.

be this non-financial income,

$$y_t(s, e, \theta) = \begin{cases} (1 - \tau_\ell)\theta^F w_t h_t e, & \text{if formal;} \\ \theta^I w_t h_t e, & \text{if informal;} \\ 0, & \text{if unemployed.} \end{cases}$$

We assume that each individual may be eligible to receive *Bolsa Família* (BF), depending on their sector and wage. We model the probability of receipt based on the data, in such a way that unemployed workers receive with a fixed probability, while informal and formal workers receive BF with a probability that decreases as income rises,

$$\mathbb{P}[i \text{ is eligible} | s, y] = \frac{\phi_s}{1 + (y/\bar{y}_s)^{1/\sigma_s}},$$

where ϕ_s is the take-up at zero earnings, $\bar{y}_s$ is the half-coverage threshold, and σ_s is the smoothness of eligibility, for each $s \in \{F, I\}$. For any income level, an informal worker is more likely to receive the benefit, $\phi_I > \phi_F$. The total number of beneficiaries is BF_t and the individual benefit is $T_{i,t}(s, y) = T_t \cdot \mathbb{1}_{i,t}\{\text{eligible}\}(s, y)$.

Households save into liquid assets $a_{i,t}$ with ex-ante real return $1 + r_t = (1 + i_{t-1})/(1 + \pi_t)$, where i_t represents the nominal interest rate and π_t denotes inflation. They also receive lump-sum transfers τ_t by the government and dividends $d_{i,t}$ from firms, which we assume to be proportional to productivity $d_{i,t} = d_t \cdot e_{i,t}$ ². Their budget constraint at time t reads:

$$c_{i,t} + a_{i,t} = (1 + r_t)a_{i,t-1} + y_{i,t}(s, e, \theta) + T_{i,t}(s, y) + d_{i,t} + \tau_t.$$

Incomplete markets impose a borrowing constraint $a_{i,t} \geq \underline{a}$. Frictions in the labor market imply formal households delegate their labor supply decisions to labor unions, and thus take worked hours as given. While, informal ones choose hours to maximize their utility, thus, for each i ,

$$\psi(h)^{1/\varphi} = \theta w e \quad \Leftrightarrow \quad h^I(e) = \left[\frac{\theta w e}{\psi} \right]^\varphi.$$

Such as Esteban-Pretel and Kitao (2021) and Finamor (2025), given a sector s , we assume that each worker faces an exogenous layoff probability δ_s , and receives an offer from s with probability π_s if employed or π_s^U if unemployed. The household accepts the offer with the higher value, smoothed by additive idiosyncratic preference shocks explained later.

²Dividends are proportional to productivity in order to increase inequality. We aim to perform robustness tests with other specifications.

The Bellman Equation for each individual i in sector $s \in \{F, I\}$ is

$$\begin{aligned}
V_t^s(a, e, \beta, \theta) = \max_{\tilde{c}, a'} & \left\{ \frac{\tilde{c}^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} + \beta \mathbb{E}_t \left[\delta_s V_{t+1}^U(.) + (1-\delta_s) \left(\right. \right. \right. \\
& \pi_F (1-\pi_I) \max \{ V_{t+1}^s(., \theta), \mathbb{E}_\theta V_{t+1}^F(., \theta') \} \\
& + (1-\pi_F) \pi_I \max \{ V_{t+1}^s(., \theta), \mathbb{E}_\theta V_{t+1}^I(., \theta') \} \\
& + \pi_F \pi_I \max \{ V_{t+1}^s(., \theta), \mathbb{E}_\theta V_{t+1}^I(., \theta'), \mathbb{E}_\theta V_{t+1}^F(., \theta') \} \\
& \left. \left. \left. + (1-\pi_F)(1-\pi_I) V_{t+1}^s(., \theta) \right) \right] \right\} \\
\text{s.t.} \quad & c + a' = (1+r_t)a + y_t(s, e, \theta) + d_{i,t} + \tau_t \\
& a' \in [\underline{a}, \infty), \quad \tilde{c} = c - v(h),
\end{aligned}$$

and for each unemployed worker is

$$\begin{aligned}
V_t^U(a, \beta) = \max_{c, a'} & \left\{ \frac{c^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} + \beta \mathbb{E}_t \left[(1-\pi_F)(1-\pi_I) V_{t+1}^U(.) \right. \right. \\
& + \pi_F (1-\pi_I) \max \{ V_{t+1}^U(.), \mathbb{E}_\theta V_{t+1}^F(.) \} \\
& + (1-\pi_F) \pi_I \max \{ V_{t+1}^U(.), \mathbb{E}_\theta V_{t+1}^I(.) \} \\
& \left. \left. + \pi_F \pi_I \max \{ V_{t+1}^U(.), \mathbb{E}_\theta V_{t+1}^I(.), \mathbb{E}_\theta V_{t+1}^F(.) \} \right] \right\} \\
\text{s.t.} \quad & c + a' = (1+r_t)a + y_t(U) + d_{i,t} + \tau_t \\
& a' \in [\underline{a}, \infty).
\end{aligned}$$

Formal workers maximize their utility taking hours as given, so their labor disutility is internalized at the union level, not by individual households. However, informal hours are chosen and therefore included in the Euler Equation. Given their decision, their consumption-equivalent income is

$$w^I e h_t - v(h_t) = w^I e h_t - \psi \frac{h_{i,t} h_{i,t}^{1/\varphi}}{1 + \frac{1}{\varphi}} = \frac{w^I e h_t}{1 + \varphi}.$$

As in Caliendo, Dvorkin, and Parro (2019), before deciding, the households receive an offer $A \in \{F, I, FI\}$ and draws a vector $\{\varepsilon^j\}$ over the options $j \in \mathcal{S}$, with scale σ_S ³, and picks the option j maximizing $V^j + \varepsilon^j$. This delivers a closed-form expected value,

$$\mathbb{P}^A(j) = \frac{\exp(V^j/\sigma_S)}{\sum_{k \in \mathcal{S}} \exp(V^k/\sigma_S)}, \quad \mathbb{E} \left[\max_{j \in \mathcal{S}} \{V^j + \varepsilon^j\} \right] = \sigma_S \ln \sum_{j \in \mathcal{S}} \exp(V^j/\sigma_S)^4.$$

³ σ_S governs the smoothness of the decision. As $\sigma_S \rightarrow 0$ it collapses to the deterministic max.

⁴We use a standard linear weighted approximation for the log-exp sum.

These shocks render the transition a differentiable function, which allows it to be linearized when computing the sequence-space Jacobians, as in Iskhakov et al. (2017). Therefore, the sector transition matrix $\Pi_t^s = \mathbb{P}(s'|s)$ is given by

$$\Pi_t^s = \begin{bmatrix} (1 - \delta_F)\phi_F(F) & (1 - \delta_F)\phi_F(I) & \delta_F \\ (1 - \delta_I)\phi_I(F) & (1 - \delta_I)\phi_I(I) & \delta_I \\ \phi_U(F) & \phi_U(I) & \phi_U(U) \end{bmatrix},$$

where $\phi_s(s')$ is the probability of accepting an offer s' given that the worker is in sector s ,

$$\begin{aligned} \phi_F(I) &= \pi_F \pi_I \mathbb{P}^{FI}(I) + (1 - \pi_F) \pi_I \mathbb{P}^I(I), & \phi_F(F) &= 1 - \phi_F(I), \\ \phi_I(F) &= \pi_F \pi_I \mathbb{P}^{FI}(F) + \pi_F (1 - \pi_I) \mathbb{P}^F(F), & \phi_I(I) &= 1 - \phi_I(F), \\ \phi_U(F) &= \pi_F \pi_I \mathbb{P}^{FI}(F) + \pi_F (1 - \pi_I) \mathbb{P}^F(F), \\ \phi_U(I) &= \pi_F \pi_I \mathbb{P}^{FI}(I) + (1 - \pi_F) \pi_I \mathbb{P}^I(I), & \phi_U(U) &= 1 - \phi_U(F) - \phi_U(I). \end{aligned}$$

We denominate the steady-state size of each sector $s \in \{F, I, U\}$ as α_s , such that $\alpha_F + \alpha_I + \alpha_U = 1$. Let $\Lambda_t(s, \beta, e, a)$ be the household distribution over a given period of time.

Firms

In the firm side, there is a representative final-good producer that aggregates a continuum of $j \in [0, 1]$ intermediate inputs produced by formal firms,

$$Y_t = \left(\int_0^1 y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

where $\varepsilon > 0$ is a constant substitution elasticity (CES).

Each intermediate good j is produced by a monopolistic competitive firm $y_{j,t} = Z_t l_{j,t}$. They hire labor at real wage w_t and set a constant markup of $\mu = \frac{\varepsilon}{\varepsilon-1}$. Hence, $P_t = \mu Z_t W_t$, and real wage equals $w_t = Z_t / \mu$ at every date in steady-state.

We assume sticky prices following Rotemberg (1982), standard in Neo-Keynesian's literature, with a quadratic adjustment cost

$$\Psi_t(\pi_t) = \frac{\mu}{\mu - 1} \frac{1}{2\kappa} [\log(1 + \pi_t)]^2 Y_t.$$

Firms' dividends are taxed at the same rate τ_ℓ as labor income, so $d_t = (1 - \tau_\ell)(Y_t - w_t L_t) - \Psi_t(\pi_t)$, where $L_t = \mathbb{E}[l_{j,t}]$ is the aggregate formal labor. Such as Hagedorn, Manovskii, and Mitman (2019), the Phillips curve for price inflation is given by

$$\ln(1 + \pi_t) = \kappa \left[\frac{w_t}{Z_t} - \frac{1}{\mu} \right] + \frac{\ln(1 + \pi_{t+1})}{1 + r_{t+1}} \cdot \frac{Y_{t+1}}{Y_t},$$

Additionally, there is a representative informal firm operating in a perfectly competitive market and producing under constant returns to scale, so $Y_t^I = w_t L_t^I$.

Unions

Nominal wages are assumed to be sticky as in Erceg, Henderson, and Levin (2000), where unions set nominal wages to maximize agent utility subject to Rotemberg (1982)'s adjustment costs. We assume that unions allocate all formal labor hours uniformly across agents, so $h_{i,t} = h_t$ (Auclert, Rognlie, and Straub 2025).

Also, as in Hagedorn, Manovskii, and Mitman (2019), union's objective is to maximize the utility of an agent with average consumption $\bar{C}_t = \int_0^1 c_{it} di$, discounted by the average discount factor $\bar{\beta} = \lambda_I \beta_L + (1 - \lambda_I) \beta_H$ (Auclert, Rognlie, Souchier, et al. 2021). This results in a Phillips curve for wage inflation $\pi_t^w = (1 + \pi_t)w_t/w_{t-1}$ given by

$$\ln(1 + \pi_t^w) = \kappa_w \left[\psi \cdot h_t^{1/\varphi} \cdot \bar{C}_t^{1/\gamma} - \frac{(1 - \tau_\ell)w_t L_t}{\mu_w h_t F_t} \right] + \bar{\beta} \ln(1 + \pi_{t+1}^w).$$

where $L_t/(h_t F_t) = \mathbb{E}(e_{i,t}|F)$, since the union's members are the formal workers only.

The Phillips Curve calculations for price and wage are found in the Appendix.

Government

Monetary Policy follows a standard lagged Taylor Rule, without microfoundations and only with an inflation target, $i_t = r_{t-1}^* + \phi \pi_{t-1}$, where r^* is the neutral real interest rate, and $\phi > 1$ (Taylor 1993).

In the governmental sphere, we have a government that collects taxes, makes transfers, and incurs debt; therefore, the government's budget constraint is characterized by

$$\tau_t + T_t \cdot \text{BF}_t + (1 + r_t)B_{t-1} = B_t + \tau_\ell Y_t.$$

We consider two types of government policy, a revenue-neutral and a deficit-financed. The policymaker chooses a path of the conditional transfers and adjusts transfers or debt such that the budget is balanced at each period.

The revenue-neutral policy sets a path of T_t and adjusts transfers such that

$$\tau_t = \tau_\ell Y_t - T_t \cdot \text{BF}_t - r_t \bar{B},$$

which means that the fiscal policy is passive and the monetary policy is active. The nominal interest rate adjust at least one-for-one to changes in inflation.

However, governments normally finance themselves by increasing deficit and issuing new debt. The deficit-financed policy sets a path of T_t and τ_t and adjusts debt, keeping the nominal interest rate fixed $i_t = \bar{i}$. We assume that the government repays its debt in

the long run by reducing transfer (when $\tau_t > 0$) or increasing taxes (when $\tau_t < 0$), with elasticity ϕ_B ,

$$\tau_t = \bar{\tau} - \phi_B \cdot (B_{t-1} - \bar{B}).$$

Or, in another words, using Auclert, Rognlie, and Straub (2025) notation, the debt persistency is defined by

$$dB_t = (1 + r_t)dB_{t-1} + d\tau_t = (1 + r - \phi_B)dB_{t-1} = \rho_B dB_{t-1},$$

where ρ_B is the debt persistency, which we calibrate so that debt is paid off on average within 10 year, so $\rho_B = 0.975$ and $\phi_B = 0.047$.

This situation leads to larger multipliers and great inequality (Auclert, Rognlie, and Straub 2023). In the presence of nominal rigidity and non- Ricardian consumer behavior, Angeletos, Lian, and Wolf (2024) show that fiscal deficits of this sort self finance, so $\lim_{t \rightarrow \infty} B_t = \bar{B}$.

Market Clearing

Finally, the asset, labor, and goods markets reach equilibrium (Market Clearing Conditions). First, the asset market reaches equilibrium when household savings equal government bonds

$$A_t = \int a_{i,t} d\Lambda_t(s, \beta, e, a) = B_t;$$

the labor market clears when the share of employment in the workforce equals the firm's demand for labor

$$\begin{aligned} L_t &= \int \mathbb{1}\{s_{i,t} = F\} \cdot \theta_{i,t}^F \cdot e_{i,t} \cdot h_{i,t} d\Lambda_t(s, \beta, e, a); \\ L_t^I &= \int \mathbb{1}\{s_{i,t} = I\} \cdot \theta_{i,t}^I \cdot e_{i,t} \cdot h_{i,t} d\Lambda_t(s, \beta, e, a); \end{aligned}$$

and the goods market clear when $Y + Y^I = C + \Psi_t$. We set the goods market to clear by Walras' Law. Lastly, the BF benefit is given to all eligible households,

$$\text{BF}_t = U_t + \int \mathbb{1}\{s_{i,t} = I, e_{i,t} < e^*\} d\Lambda(s, \beta, e, a).$$

2. Calibration

Data

To calibrate this model, we use Brazilian data and standard parameters in HANK Literature. Most of the data refers to their average values in 2025.

The labor market moments were obtained in *Pesquisa Nacional por Amostra de Domicílio Contínua* (PNAD-C), an annual survey reporting formality status, wages,

weekly worked hours, etc. This panel is interesting to match the sector transitions and the worked hours and wages, for both formal and informal households. However, since it is a survey, it might have under-reporting of informal households.

Some general macroeconomic measures were also used, such as the Government Debt (in the Brazilian National Treasury), the interest rates, and the inflation, available in the *Índice Brasileiro de Geografia e Estatística* (IBGE). The Gini coefficient of wealth was obtained from the 2025 Global Wealth Report, from UBS, and the Wealth Inequality Statistics were obtained from the 2024 Wealth Inequality Database⁵. *Bolsa Família* data is available in *Cadastro Único* (CadÚnico), in VIS Data⁶, a platform with several statistics about the benefit.

Additionally, Annual PNAD-C has information about whether a individual is beneficiary of the benefit. We identify each household and consider every individual living in a beneficiary household as receiving the benefit. Among those, we calculate the mean reported benefit T and the mean share of beneficiaries.

Table 1: Bolsa Família 2025: Statistics

	PNAD	VisData	Ratio
Households (M)	13.64	18.71	1.37
Individuals (M)	46.60	48.92	1.05
People/HH	3.42	2.61	0.77
Value (R\$/month)	669.55	710.65	1.06

Table 2: Bolsa Família 2025: Coverage

	%
BF / Labor Force	16.1
BF in Formal	7.2
BF in Informal	23.9
BF in Unemployed	33.6
BF in Non-Participating	21.8
BF HH with a Worker	81.6
Spending / Wage Bill	2.2
$T/\mathbb{E}(y^F)$	10.6

The rule to receive BF is based on a R\$ 218 threshold for the family income per capita, therefore there are some formal individuals receiving the benefit. Also, CadÚnico normally delay to remove the benefit for non-eligible individuals, which can explain why some formal individuals receive BF.

In order to construct quarter-to-quarter transition probabilities across the three sectors $s \in \{F, I, U\}$, we reconstruct the PNAD-C rotating panel using Osório (2022), which

⁵Available in [WID World](#).

⁶Available in [VIS DATA 3](#) and [Cadastro Único](#).

estimates the dwelling and individual identifiers in the survey. The estimation of the pandemic years is not reliable.

The formality status is defined as (i) formal workers have signed work card, are militaries, or employers, and (ii) informal workers do not have a signed work card, are self-employed, or unpaid family. We only consider household in the labor force, dropping those who are non-participating, retired, or students, to keep the empirical states aligned with the model's.

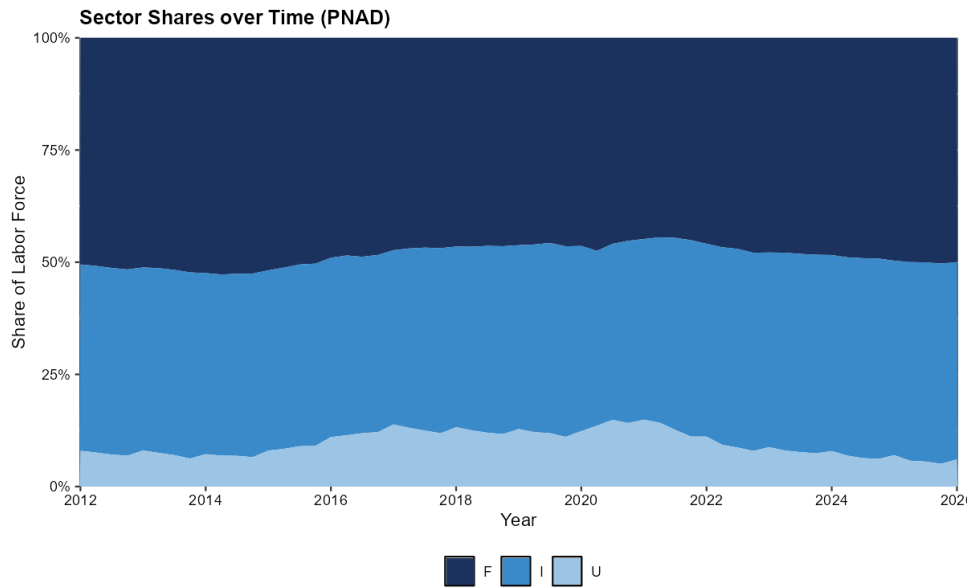


Figure 1: Share of Formal, Informal, and Unemployed Households

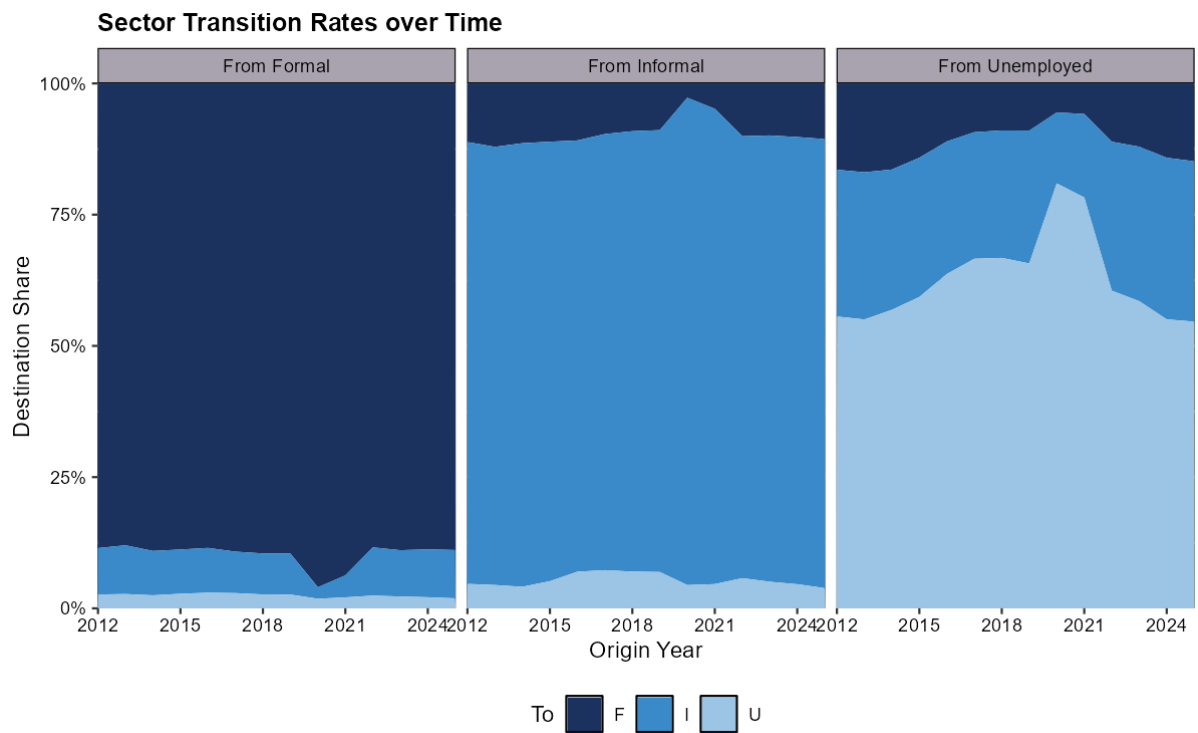


Figure 2: Sector Transitions over time

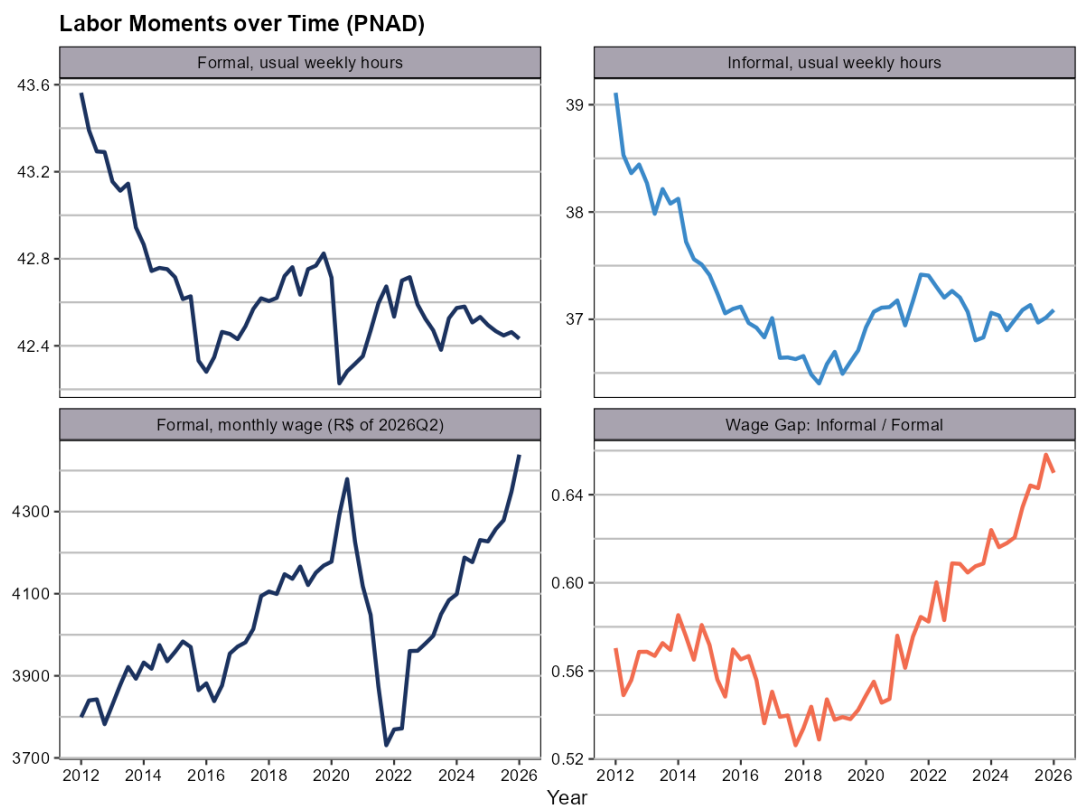


Figure 3: Wage and Worked Hours for each Sector

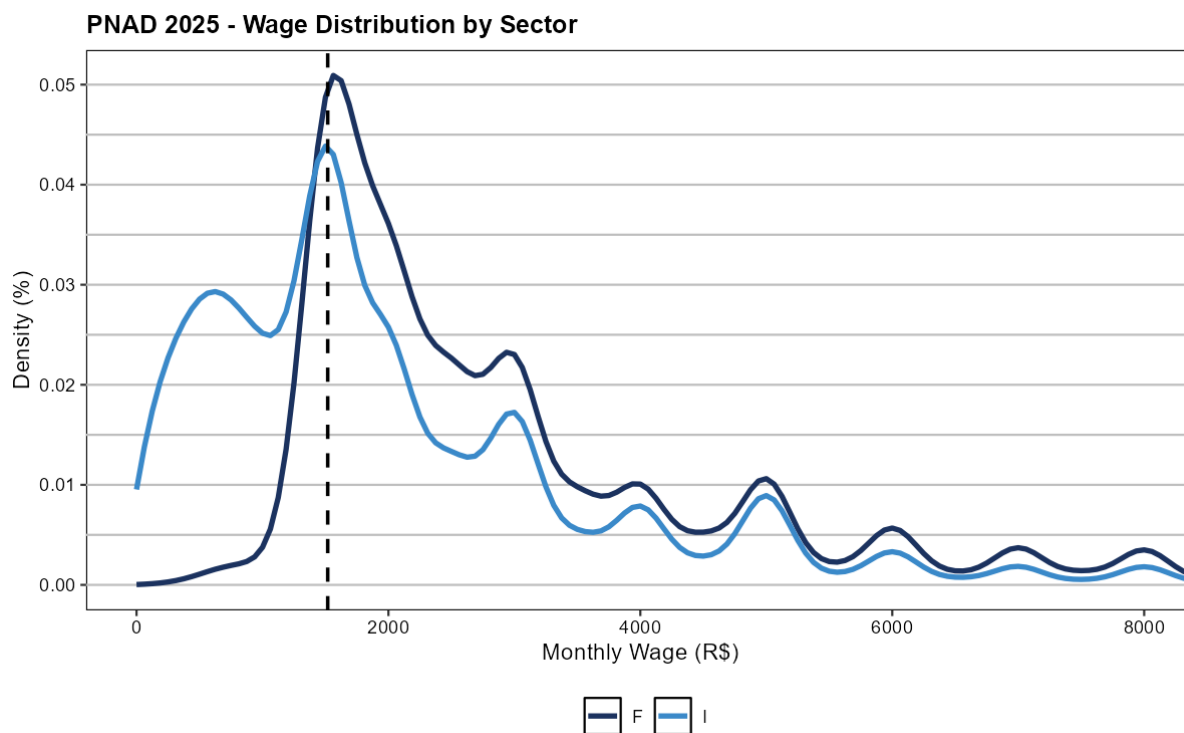


Figure 4: Wage Distribution for each Sector

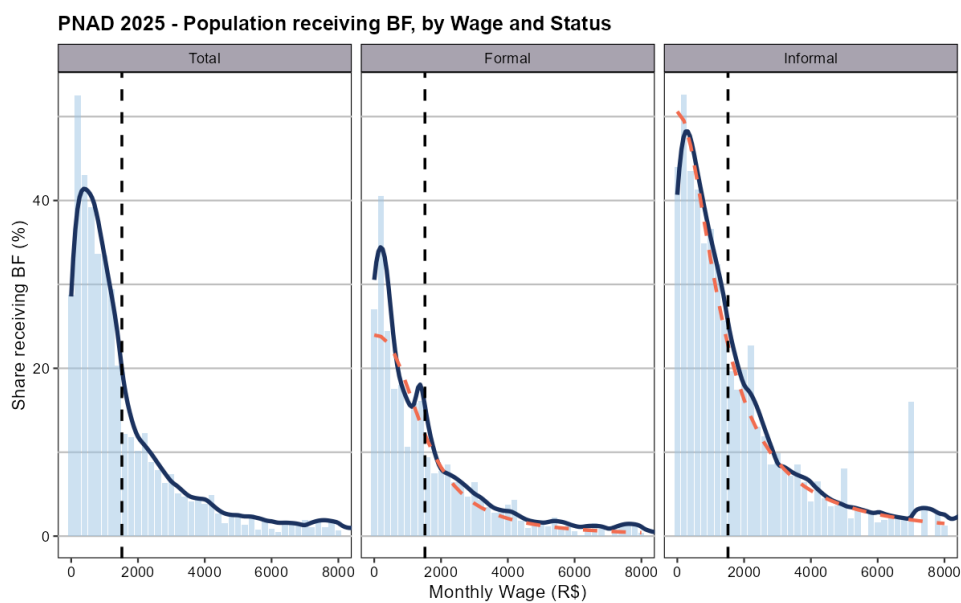


Figure 5: Share of Households Receiving Benefit by Wage Level

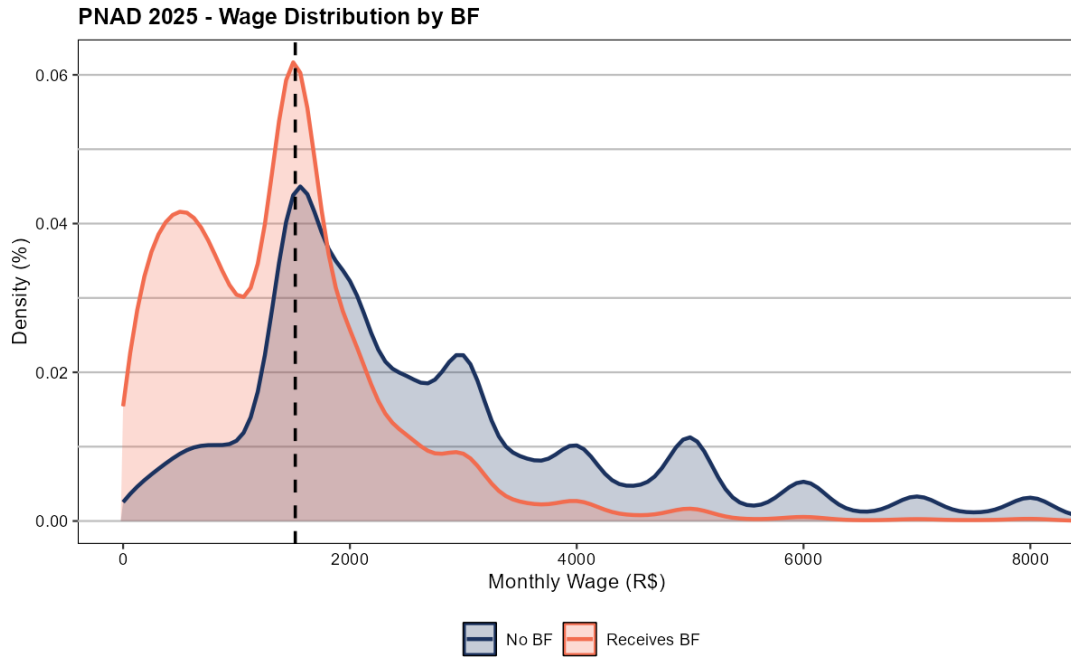


Figure 6: Wage Distribution by Benefit Status

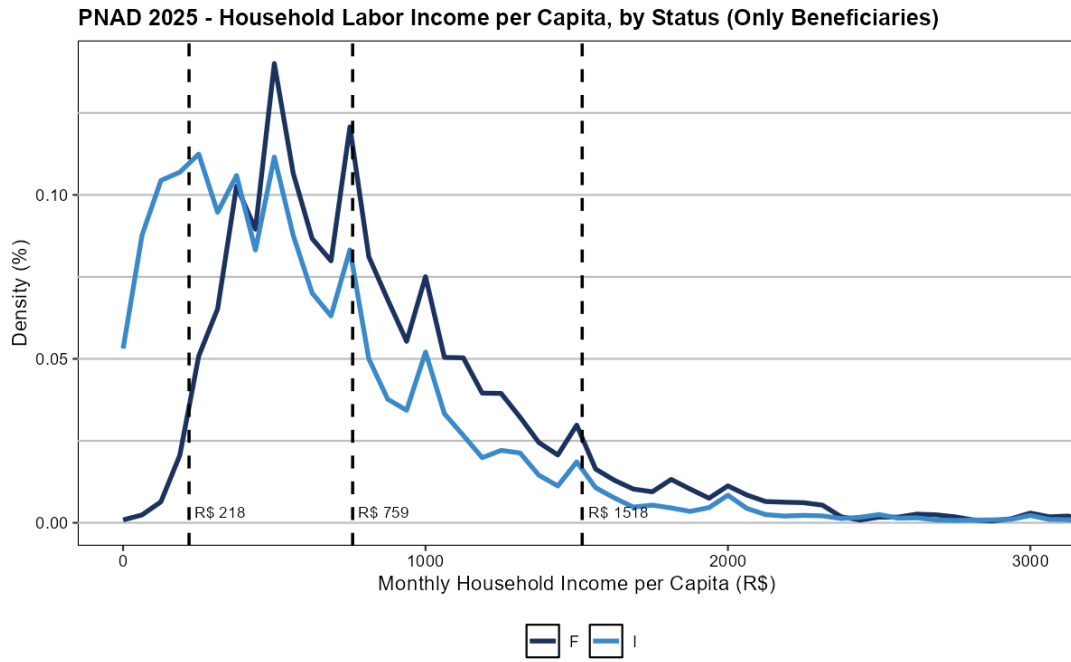


Figure 7: Familiar Wage Distribution by Sector, for Beneficiaries

Methods

The main computational methods are the Sequence-Space Jacobian (SSJ) (Auclert, Bardóczy, et al. 2021), and the Endogenous Grid Method (EGM) (Carroll 2006). The

EGM was combined with the discrete choice among sectors using Iskhakov et al. (2017), with taste shocks.

The central idea of SSJ is to summarize the entire model by a matrix equation $\mathbf{F}_t(\mathbf{X}, \mathbf{Z}) = \mathbf{0}$, where $\mathbf{X}$ are the endogenous variables and $\mathbf{Z}$ are the exogenous ones. The Implicit Function Theorem allows us to differentiate this system and obtain General Equilibrium Jacobians, written as

$$d\mathbf{X} = \mathbf{G} d\mathbf{Z} = -\mathbf{F}_{\mathbf{X}}^{-1} \mathbf{F}_{\mathbf{Z}} d\mathbf{Z}.$$

For each shock in *Bolsa Família*, we can compute the Jacobian $dC = \mathbf{G}_{C,T} dT$ of the change in the conditional transfer on household consumption. The Jacobian is computed by perturbing the exogenous variable T and solving the model for the new equilibrium, then calculating the change in consumption C .

Further, we can compute the consumption response to a monetary shock $dC = \mathbf{G}_{C,i} di$ in the presence of the transfer, and compare it to the counterfactual. We measure the stabilization effect of the transfer as the difference between the response in the economy with the program and in the $T = 0$ counterfactual,

$$\text{Stab}(T) = dC \Big|_{T>0} - dC \Big|_{T=0},$$

read either as an impulse response or as the induced change in the volatility of consumption $\Delta \text{Var}(C_t)$, a program that smooths fluctuations delivers $\Delta \text{Var}(C_t) < 0$. This effect can be decomposed in a insurance and a composition channel,

$$d\mathbf{X} = d\mathbf{X}^{\text{ins}} + (d\mathbf{X} - d\mathbf{X}^{\text{ins}}) = \mathbf{G}^{\text{ins}} d\mathbf{Z} + (\mathbf{G} - \mathbf{G}^{\text{ins}}) d\mathbf{Z}.$$

The sectoral block of the transition matrix Π_t is endogenous and depends on the continuation values, $\Pi_t = \Pi(\mathbf{V}_t, \Lambda_t)$, where $\mathbf{V}_t = (V_t^F, V_t^I, V_t^U)$. The household's distribution evolves by combining the asset policy with the Markov transition over the states, $\Lambda_{t+1} = \mathcal{T}(\Pi_t, a_t(s, \theta, \beta, e))$. To solve the steady-state distribution, we iterate until Π_t converges to its fixed point, analogue to Aguirregabiria and Mira (2007).

The insurance channel is captured by the frozen- Π . Holding sectoral sorting fixed, the transfer relaxes the budget of unemployed and low-income informal households, and, since these are high-MPC, it dampens the aggregate response (Auclert, Rognlie, and Straub 2024).

The composition channel is captured by the moving- Π Jacobian, because the CCT raises the value of informality relative to formal work, so workers are tilted toward the informal sector, which is riskier. The insurance term is stabilizing, while the composition is potentially destabilizing. The sign and magnitude of the net effect are the quantitative answer to our question.

Letting the transition respond, $d\Pi_t = \Pi_{\mathbf{V}} d\mathbf{V}_t$, yields the general equilibrium Jacobian $\mathbf{G}$. This dynamic is obtained as the dynamic analogue of the steady-state: given a shock path, we iterate so that households' sectoral choices are consistent with the values those very choices generate along the transition (Auclert, Bardóczy, et al. 2021).

Calibration

Table 3: External Calibration

Description		Value	Source
<i>Households and Labor Market</i>			
γ	EIS	0.5	Standard in Literature
φ	Informal Frisch Elasticity	0.200	Standard in Literature
ψ	Disutility of Labor	1.249	Normalization: $h_F = 1.0$
β_H	Patient's Discount Factor	0.958	$A/Y = B/Y = 78.6\%$
δ_F	Job-Loss Probability to Formal	0.009	PNAD-C Data (Osório 2022)
δ_I	Job-Loss Probability to Informal	0.038	PNAD-C Data (Osório 2022)
<i>Firms</i>			
w	Average Wage	1.0	Normalization, Reference
μ, μ_w	Price Markup	1.11	Basu and Fernald (1997)
κ	Price-NKPC Slope	0.10	Kaplan, Moll, and Violante (2018)
κ_w	Wage-NKPC slope	0.10	1 year of Wage Rigidity
<i>Government and Monetary Policy</i>			
r^*	Neutral Real Interest Rate	1.2%	5% annual average
τ_ℓ	Labor Income Tax Rate	13.0%	Tax Payroll for Average Worker
ϕ	Taylor Rule Coefficient	1.5	Standard (Taylor 1993)
B/Y	Government Debt-to-GDP Ratio	78.6%	Brazilian National Treasure
ϕ_B	Sensitivity to Pay the Debt	0.047	B half-life of 5 years
<i>Bolsa Família</i>			
$BF \cdot T/W$	Bolsa Família Spending (%Wage Bill)	2.2%	CadÚnico, PNAD-C
ϕ_F, ϕ_I	Take-up at Zero Earnings	0.24, 0.51	PNAD-C
$\bar{y}_F, \bar{y}_I$	Half-Coverage Threshold	0.36, 0.32	PNAD-C
σ_F, σ_I	Smoothness of Eligibility	0.41, 0.51	PNAD-C

Each model period stands for one quarter and the computational time horizon is 100 quarters (or 25 years). We externally calibrate β_H to balance the asset market and π_F , π_I , π_{UF} , π_{UI} to match the sectoral probabilities. Also, we internally calibrate parameters using the Simulated Method of Moments (SMM) by a Global Search on a coarse grid, followed by a local polish using the Nelder-Mead algorithm. The external calibration is shown in table 3 and the endogenous calibration is in 4.

The idiosyncratic productivity follows a log-normal $AR(1)$ process, $\ln e' = \rho_e \ln e + \varepsilon^e$, with $\varepsilon^e \sim N(0, \sigma_e^2)$. This is approximated by a finite Markov chain with $N_e = 15$ grid points (Rouwenhorst 1995). We internally calibrate ρ_e and σ_e to match the autocorrelation of wages in PNAD-C $\rho_1^y := \text{Corr}(\ln y_t, \ln y_{t-1})$ and the observed wage distribution.

Table 4: Endogenous Calibration

Description		Value	Target	Model	Data	Source
<i>Households</i>						
β_L	Impatient's Discount Factor	0.798	Top 10%	70.7%	71.9%	WID World
ω_I	Share of Impatient Agents	0.500	HtM Share	12.2%	35.0%	Standard
σ_S	Smoothness of Tastes	0.500	Δ_{IF}	0.633	0.147	PNAD-C
<i>Labor Market</i>						
π_F, π_F^U	Formal Offer Probability	24.5%, 32.0%	F	49.8%	49.8%	PNAD-C
π_I, π_I^U	Informal Offer Probability	67.4%, 63.3%	I	44.5%	44.5%	PNAD-C
ρ_e	Persistence of Productivity	0.966	ρ_1^y	0.952	0.882	PNAD-C
σ_e	Productivity Std. Dev.	0.700	q_{90}	0.731	0.637	PNAD-C
μ_I	Informal Average Productivity	-0.540	q_{50}^I	-1.918	-0.972	PNAD-C
σ_F	Formal Std. Dev. Produc.	0.480	q_{50}^F	-0.247	-0.526	PNAD-C
σ_I	Informal Std. Dev. Produc.	0.400	q_{75}^I	-1.159	-0.344	PNAD-C

We also assume a log-spaced grid for assets $\ln a \in [\ln \underline{a}, \ln \bar{a}] = [0, 500]$ with $N_a = 200$ points, sufficient to cover the top distribution, and a log-normal sector productivity grid $\ln \theta^s \sim N(\mu_s, \sigma_s^2)$, that is discretized by Equiprobable Bins with $N_\theta = 3$ nodes (Meghir, Narita, and Robin 2015).

Each household has a $q = 0.0$ probability of drawing a new discount factor (β_L, β_H) , corresponding to an average type duration of 25 years, used to capture a generation (Auclert, Rognlie, and Straub 2025). This aims to capture similar results in relation to a life-cycle overlapping generations with permanent ability shock model.

In the household preferences, elasticity of intertemporal substitution is standard in literature $\gamma = 0.5^7$. The disutility of labor $\psi = 1.249$ is calibrated to normalize formal hours $h_F = 1.0$. Finally, the preference shock smoothing parameter is set to $\sigma_S = 0.500$, so as to match the income growth of workers switching between the formal and informal sectors, $\Delta_{FI} = \mathbb{E}[\Delta \ln y_{t+1} | F \rightarrow I] = -0.096$ and $\Delta_{IF} = \mathbb{E}[\Delta \ln y_{t+1} | I \rightarrow F] = 0.147$.

The patient's discount factor $\beta_H = 0.958$, balances the asset market. We estimate the impatient's discount factor, $\beta_L = 0.798$, and the share of impatient households, $\omega_I = 0.500$, to match the Top 10% Share of Wealth (71.9%, from World Inequality Database) and the share of Hand-to-Mouth households (35.0%)⁸, letting the Wealth Gini Coefficient (0.820, from UBI Global Report) and the Top 1% Share of Wealth (39.5%) untargeted.

The layoff probabilities are equal to the transition probabilities in PNAD-C, $\delta_F = \mathbb{P}[F \rightarrow U] = 0.009$ and $\delta_I = \mathbb{P}[I \rightarrow U] = 0.038$. Further, the job offer probabilities are calibrated to match sector sizes $s \in \{F, I, U\}$ in equilibrium, as shown in Table 5. The taste preferences smoothness is governed by $\sigma_S = 0.500$.

We internally calibrate moments of θ to match five quantiles $\{0.10, 0.25, 0.50, 0.75, 0.90\}$ of the observed wage distribution for formal and informal workers, with respect to $\mathbb{E}(y^F)$,

⁷EIS $\gamma = 0.5$ is consistent with micro estimates surveyed by Valickova, Havranek, and Horvath (2015).

⁸HtM share consistent with the estimates of Costa-Junior, Gadelha, and Garcia-Cintado (2025) and Galvêas (2026).

Table 5: Quarterly Labor-Market Transitions (%)

Origin	Model			PNAD-C		
	F	I	U	F	I	U
Formal	88.81	9.32	1.87	88.81	9.32	1.87
Informal	10.64	85.61	3.75	10.64	85.61	3.75
Unemployed	14.93	31.25	53.82	14.93	31.25	53.82
Stationary Share	49.84	44.52	5.63	49.84	44.52	5.63

which yields $\mu_I = -0.540$, $\sigma_F = 0.480$, and $\sigma_I = 0.400$, after normalizing $\mathbb{E}(\theta^F) = 1.0$.

The average wage is normalized $w = 1.0$, so that we can reference the model as a percentage of the Wage Bill. That was preferred rather than normalizing the GDP, because there is no capital in this model, therefore the wage is a better measure of the economy's size. We defined $\mu = 1.11$, which is standard in the HANK literature (Kaplan, Moll, and Violante 2018; Auclert, Rognlie, and Straub 2025)⁹, and $\kappa = \kappa_w = 0.10$, similar to Kaplan, Moll, and Violante (2018) and consistent with an average duration of 4 quarters for readjustment.

Taylor Rule is pretty standard, with $\phi = 1.5$ and $r^* = 1.2\%$, consistent with a 5% annual real interest rate for Brazilian economy (Santos and Muinhos 2024). The Government debt is set to 78.6% of yearly GDP, consistent with Brazilian National Treasury data. Also, we set $\phi_B = 0.047$ so that the half-life of the debt is 5 years, slower than Auclert, Rognlie, and Straub (2025) and consistent with Campos and Cysne (2019). Wage tax is based on the average formal worker contribution, $\tau_\ell = 13.0\%$.

We calibrated the *Bolsa Família* transfer size $T = 0.150$ to match the *Bolsa Família* expenditure as a percentage of the Wage Bill, $\text{BF} \cdot T/W$ from Brazilian economy. In order to keep the internal consistency of the model, we consider T as the average transfer received only by individuals in labor force, disconsidering children and elderly. On the other hand, we calibrated the benefit's eligibility parameters to match PNAD-C data (Figure 5), with a Weighted Non-Linear Least Square (Weighted NLS) estimation.

⁹Markup $\mu = 1.11$ is also consistent with the estimates of Basu and Fernald (1997) for US. De Loecker, Eeckhout, and Mongey (2021) show that the Brazilian markup is similar to the US.

3. Results

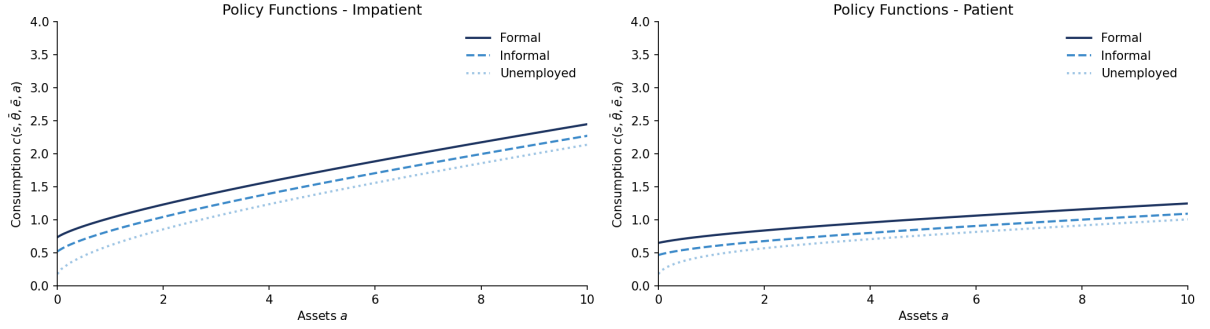


Figure 8: Steady-State Consumption Policy Function

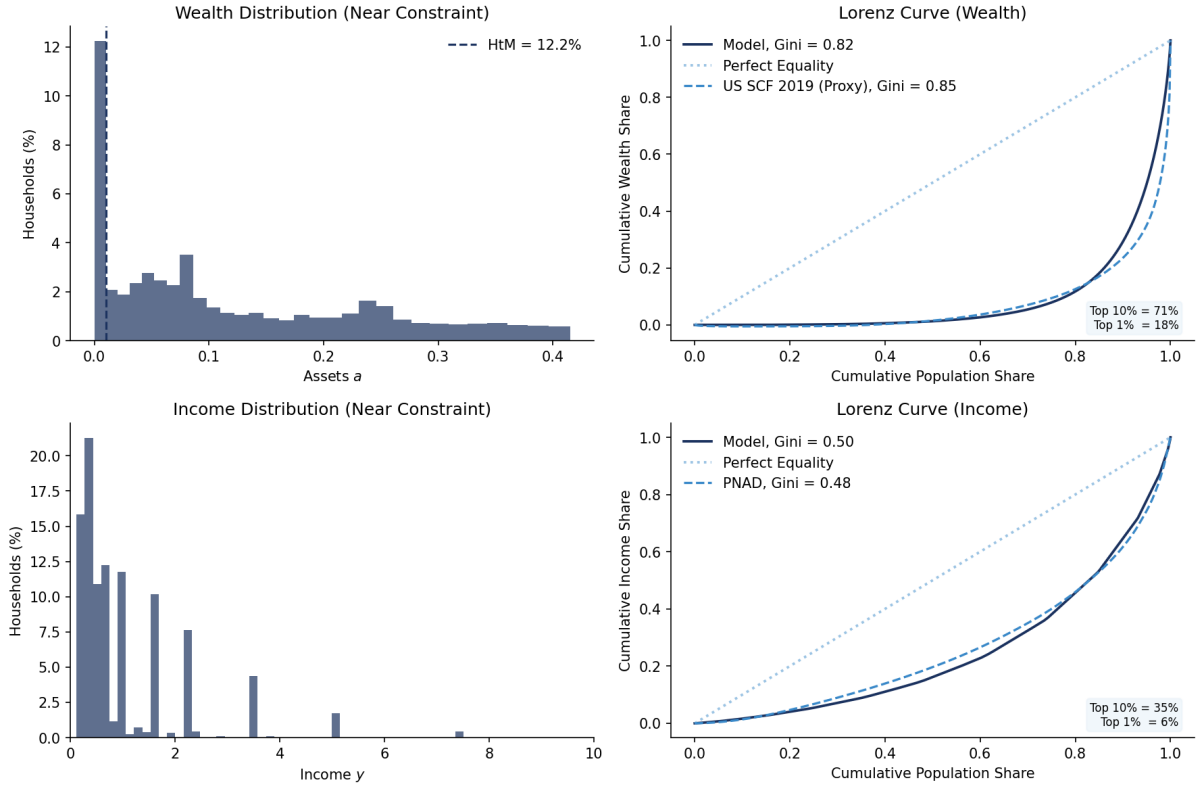


Figure 9: Wealth Distribution

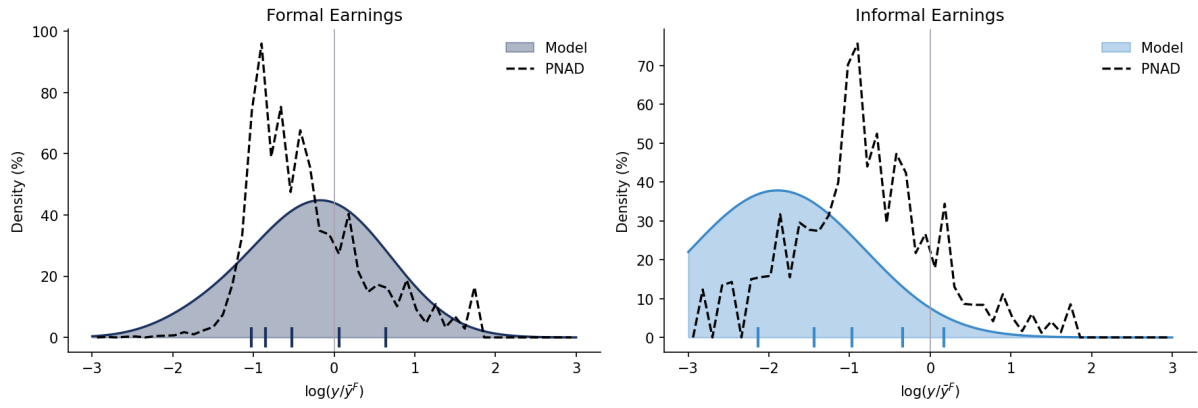


Figure 10: Income Distribution

Table 6: Steady State: Bolsa Família vs. No-Transfer Counterfactual

	With BF	No BF	Δ
Formal Share α_F	49.8%	49.9%	-0.1
Informal Share α_I	44.5%	44.0%	+0.5
Unemployment α_U	5.6%	6.0%	-0.4
Hand-to-Mouth	13.9%	9.6%	+4.3
Wealth Gini	0.802	0.783	+0.019
Consumption Gini	0.443	0.452	-0.009
Aggregate Consumption C	1.336	1.340	-0.004
BF Spending	0.027	0.000	+0.027
Welfare $\mathbb{E}[V]$	-19.534	-20.813	+1.279

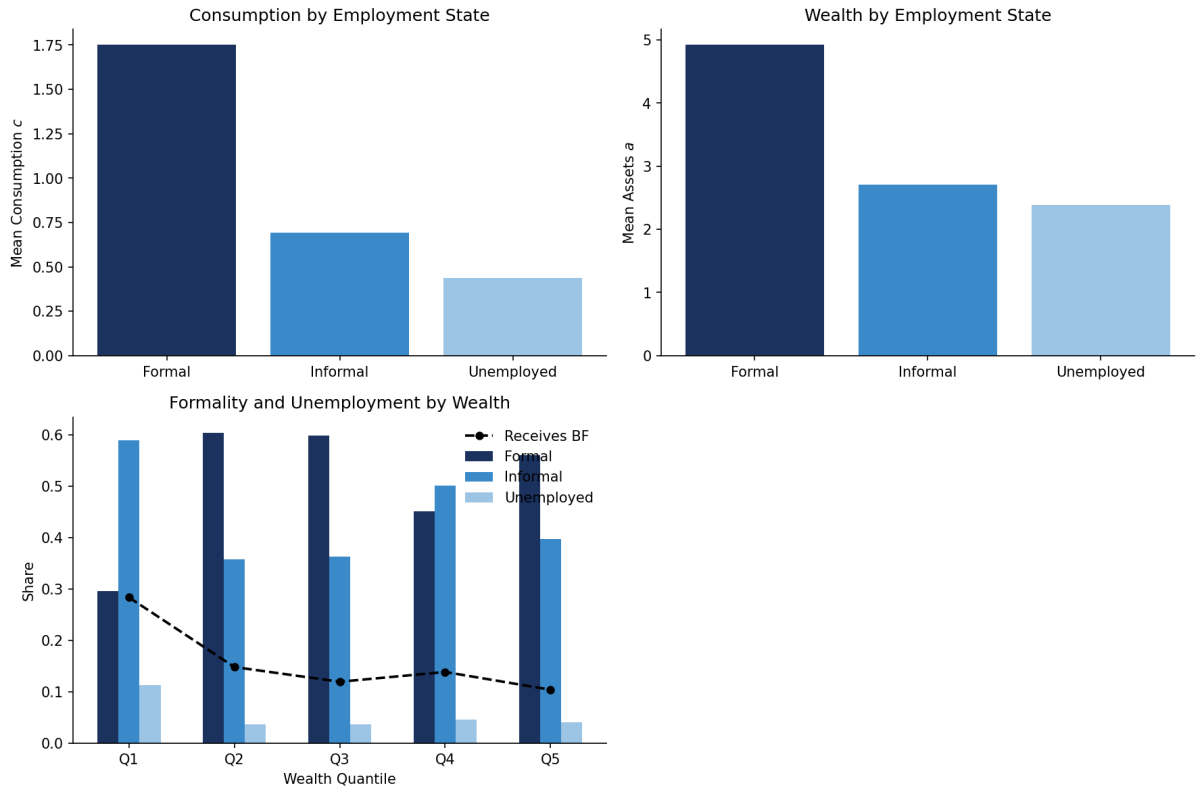


Figure 11: Steady-State Descriptive Statistics



Figure 12: Steady-State Sensitivity Analysis

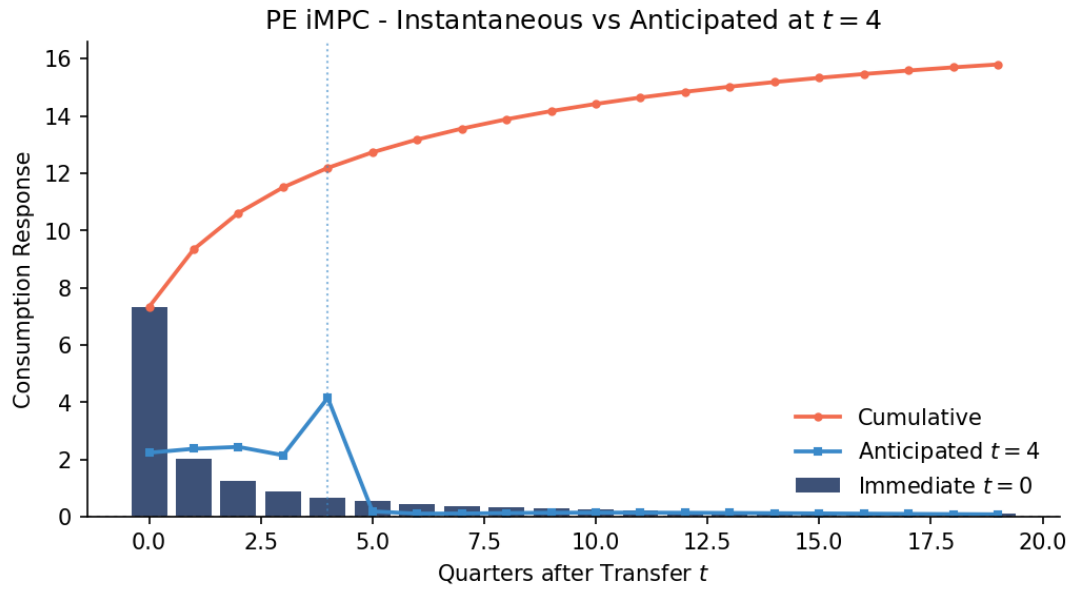


Figure 13: iMPC: Partial Equilibrium Profile

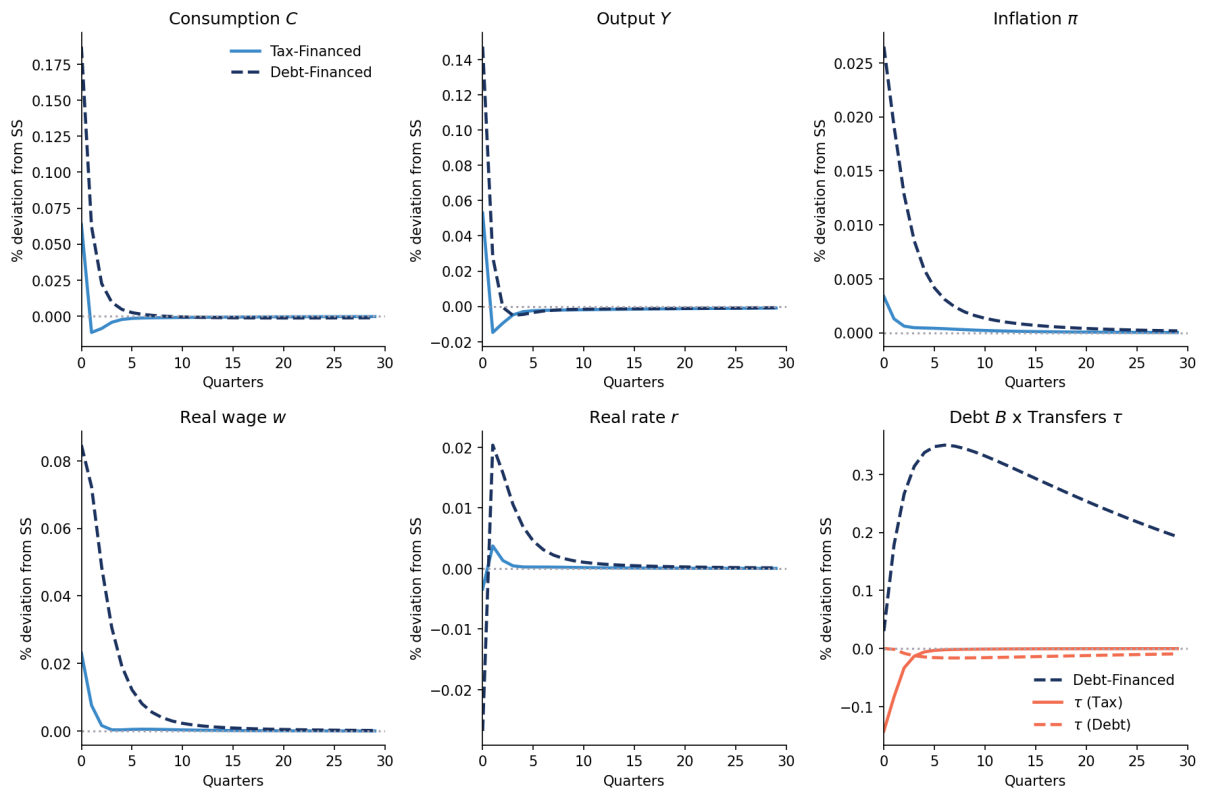


Figure 14: Response to a Shock in T by Financing Rules

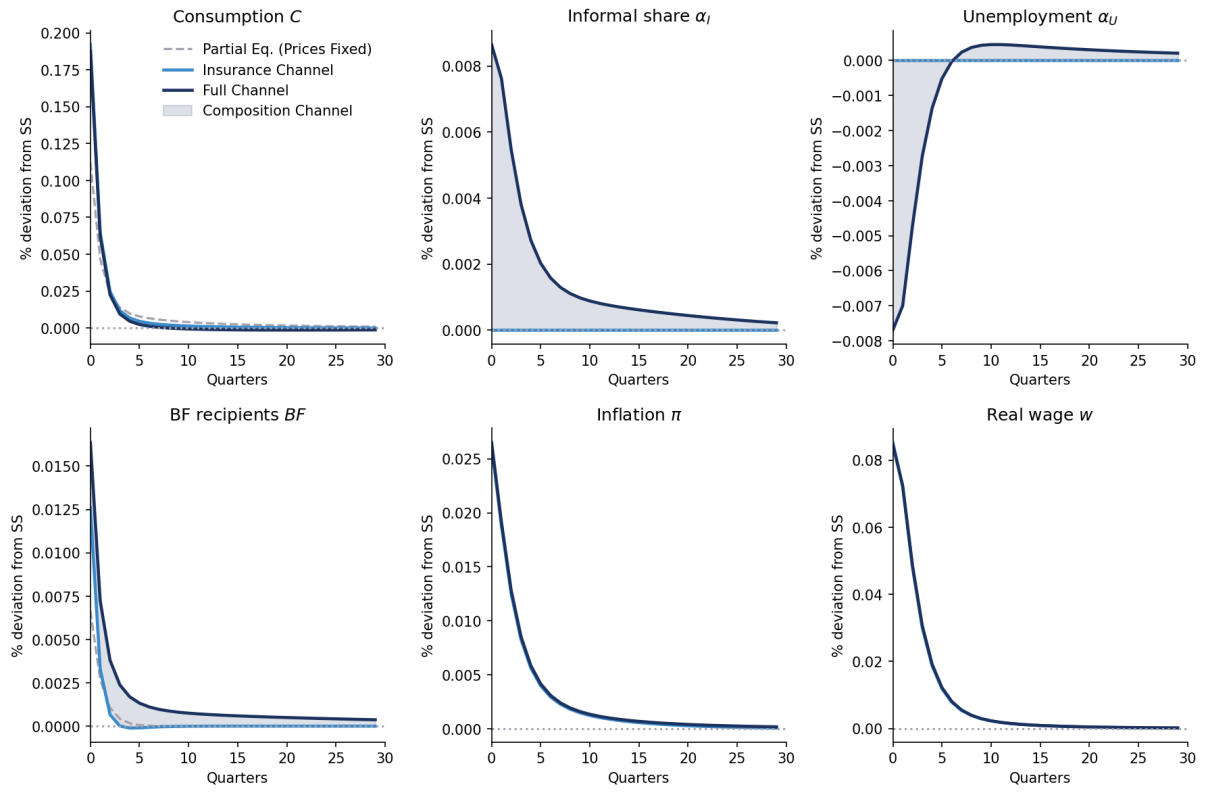


Figure 15: Response to a Shock in T by Channel

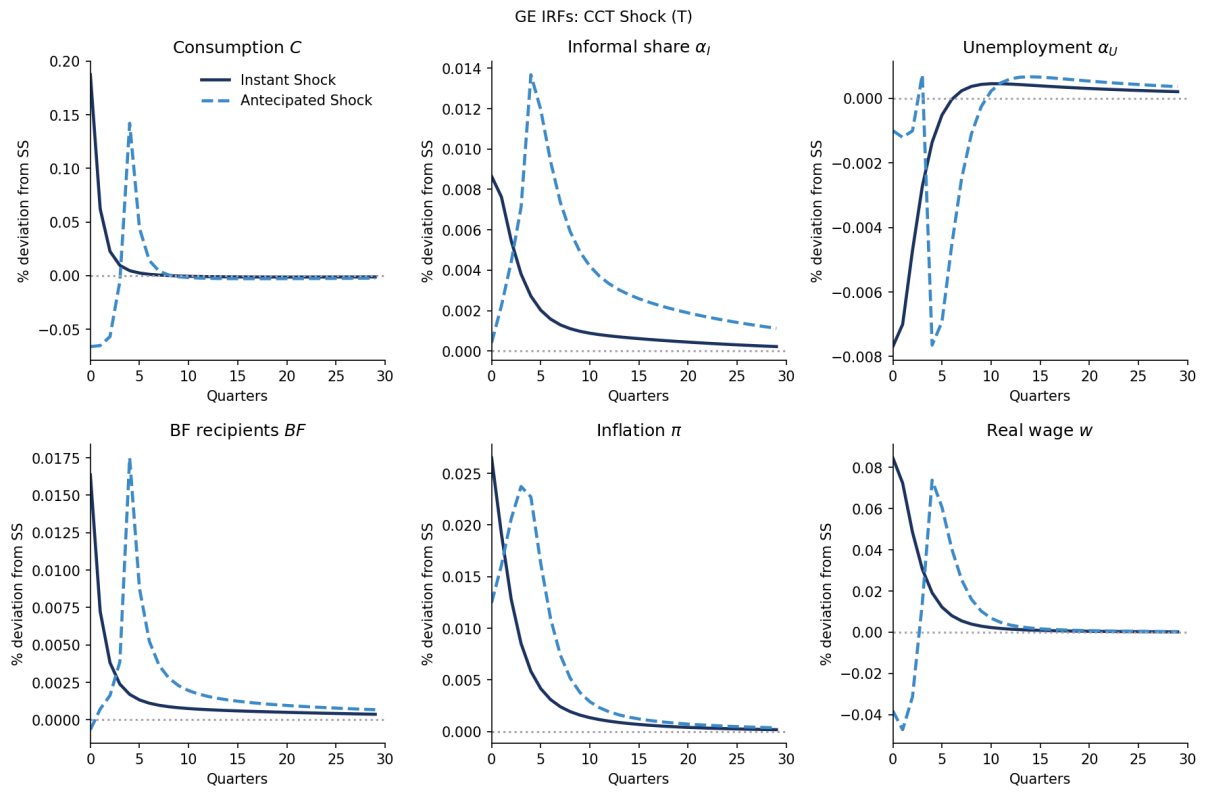


Figure 16: Response to a Shock in T : Instant vs Anticipated

Table 7: Cumulative Response (%), Shock in T				
Variable	Horizon	Insurance	Full	Composition
Consumption C	1	0.192	0.187	-0.005
	4	0.294	0.282	-0.012
	20	0.322	0.278	-0.043
	100	0.322	0.229	-0.092
Unemployment α_U	1	0.000	-0.008	-0.008
	4	-0.000	-0.022	-0.022
	20	-0.000	-0.019	-0.019
	100	-0.000	-0.012	-0.012
Inflation π	1	0.026	0.027	0.000
	4	0.066	0.067	0.001
	20	0.090	0.094	0.003
	100	0.094	0.102	0.008
Real wage w	1	0.085	0.085	-0.001
	4	0.235	0.236	0.001
	20	0.297	0.299	0.002
	100	0.301	0.305	0.004

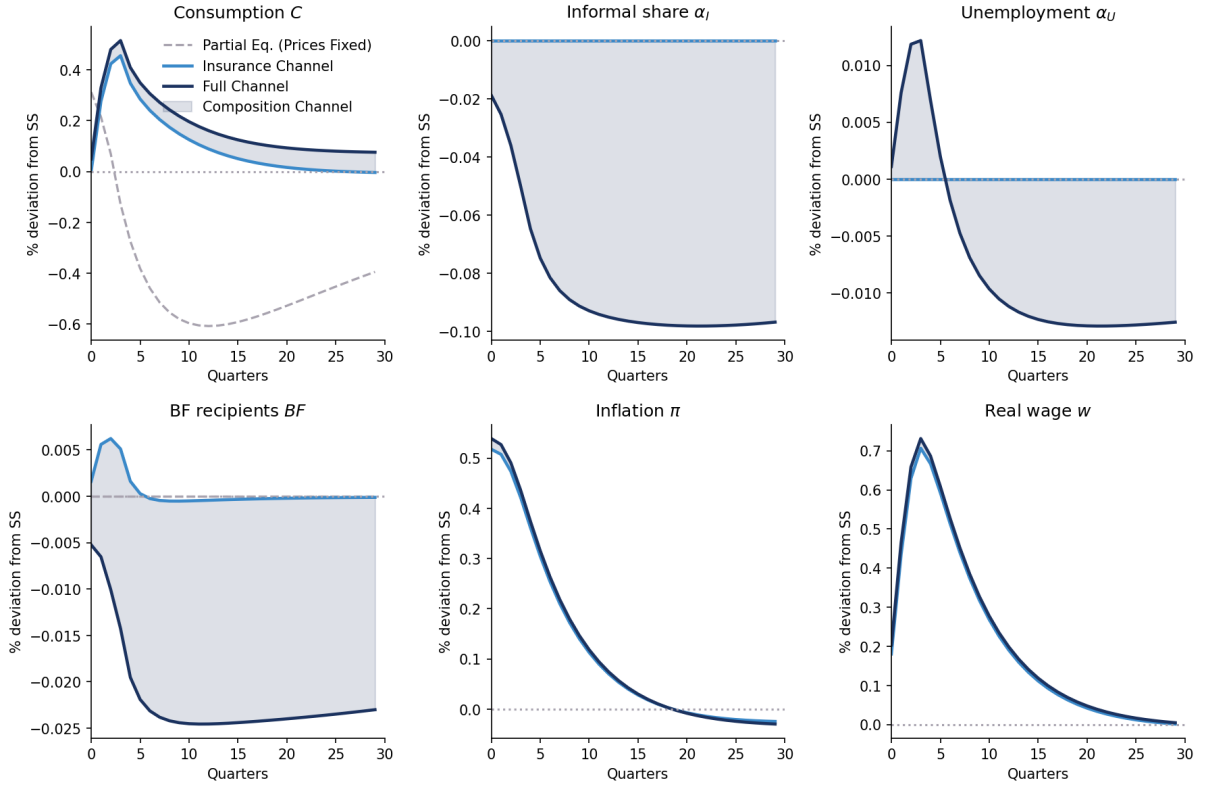


Figure 17: Response to an increase in i by Channel

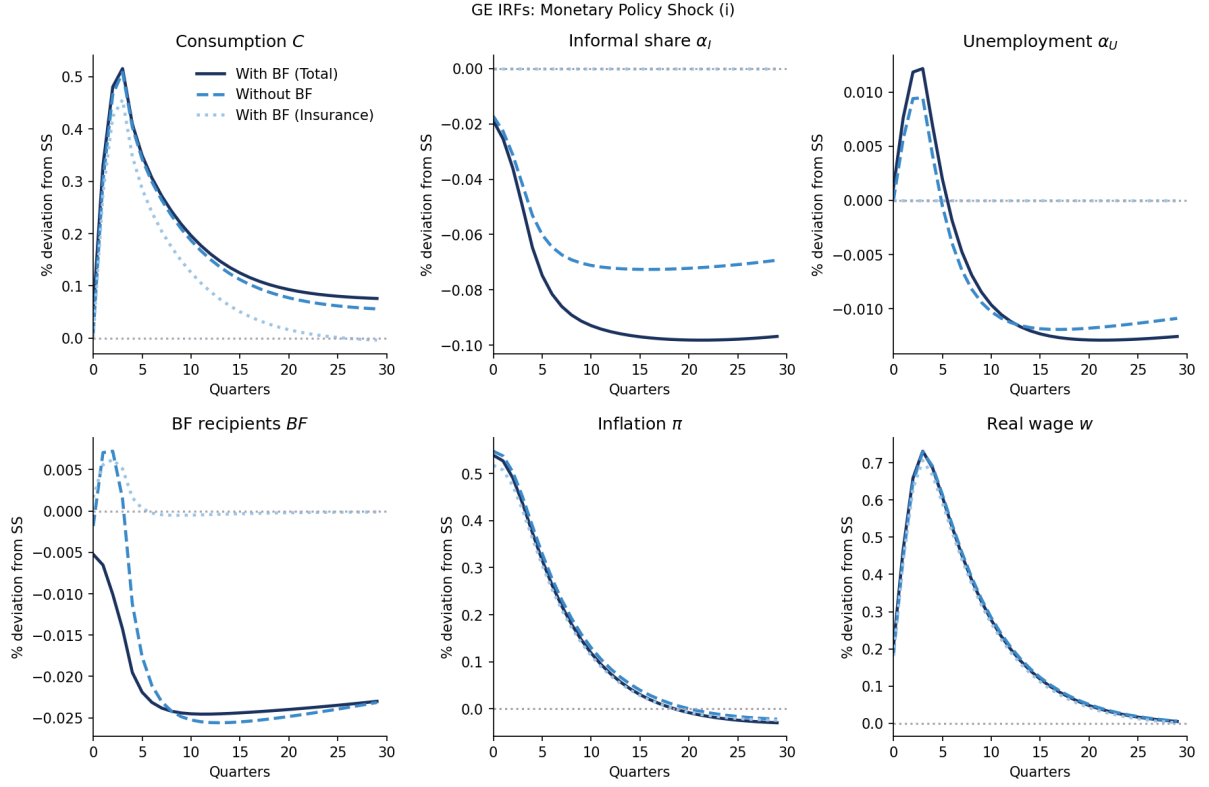


Figure 18: Response to an increase in i : With and Without CCT

Table 8: Cumulative Response (%), Shock in i

Variable	Horizon	Insurance	Full	Composition
Consumption C	1	0.002	0.050	0.048
	4	1.158	1.375	0.217
	20	3.187	4.537	1.350
	100	2.528	9.217	6.689
Unemployment α_U	1	0.000	0.001	0.001
	4	-0.000	0.033	0.033
	20	-0.000	-0.099	-0.099
	100	-0.000	-0.987	-0.987
Inflation π	1	0.518	0.539	0.021
	4	1.921	1.993	0.072
	20	3.788	3.938	0.150
	100	1.316	-1.034	-2.351
Real wage w	1	0.180	0.203	0.023
	4	1.952	2.059	0.107
	20	6.229	6.494	0.265
	100	5.817	5.751	-0.066

Robustness Tests

Lastly, in order to verify whether this results are representatives and robust, we intent to do some robustness tests changing some calibration parameters, such as γ , r^* (4% and 6%), and ϕ (2.0). This parameters might have minor implications in the results, but it is expected that the model continues to have the same results.

Also, we intent to perform other dividend specifications instead of $d_{i,t} = d_t \cdot e_{i,t}$. We also aim to change the GHH utility to a standard separable utility in the form of $u(c) - v(h)$, which must have similar results.

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Appendix

Price Phillips Curve

Let Y_t be the aggregate output of the economy, a CES of the intermediary inputs,

$$Y_t = \left(\int_0^1 y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}.$$

This firm maximizes its output subject to $\int_0^1 p_{i,j} y_{i,j} dj$. The FOC implies that the demand for the intermediate good is given by

$$y_{j,t} = \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} Y_t,$$

where P_t is the equilibrium price of the final good,

$$P_t = \left(\int_0^1 p_{j,t}^{1-\varepsilon} dj \right)^{\frac{1}{1-\varepsilon}}.$$

We assume a Rotemberg (1982) quadratic price adjustment cost given by

$$\frac{\theta_p}{2} \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right)^2 Y_t,$$

where $\theta_p = (\varepsilon - 1)/\kappa$ is the adjustment cost multiplier and $\bar{\Pi}$ the steady-state inflation.

Let $\Xi(y)$ be the intermediary firm's cost. Given last period's price $p_{j,t-1}$, the firm chooses this period's price $p_{j,t}$ to maximize the present discounted values of future profits,

$$V_t^{IF}(p_{j,t-1}) = \max_{p_{j,t}} \frac{p_{j,t}}{P_t} y_{j,t} - \Xi(y_{j,t}) - \frac{\theta_p}{2} \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right)^2 Y_t + \frac{1}{1+r_{t+1}} V_{t+1}^{IF}(p_{j,t}).$$

We can replace the demand function into this problem, so

$$V_t(p_{j,t-1}) = \max_{p_{j,t}} \frac{p_{j,t}}{P_t} \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} Y_t - \Xi(y_{j,t}) - \frac{\theta_p}{2} \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right)^2 Y_t + \frac{1}{1+r_{t+1}} V_{t+1}(p_{j,t}).$$

The FOC with respect to $p_{j,t}$ is

$$(1 - \varepsilon) \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} \frac{Y_t}{P_t} + \varepsilon m c_{j,t} - \theta_p \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right) \frac{Y_t}{p_{j,t}} + \frac{1}{1+r_{t+1}} V'_{t+1}(p_{j,t}) = 0,$$

where $m c_t = \partial \Xi / \partial y_{j,t}$, and the envelope condition gives us

$$V'_{t+1} = \theta_p \left(\frac{p_{j,t+1}}{p_{j,t}} - \bar{\Pi} \right) \frac{p_{j,t+1}}{p_{j,t}} \frac{Y_{t+1}}{p_{j,t}}.$$

Thus, taking $p_{j,t} = P_t$ in equilibrium,

$$\theta_p (\pi_t - \bar{\Pi}) \pi_t = \varepsilon \left(m c_t - \frac{\varepsilon - 1}{\varepsilon} \right) + \frac{1}{1+r_{t+1}} \theta_p (\pi_{t+1} - \bar{\Pi}) \pi_{t+1} \frac{Y_{t+1}}{Y_t},$$

where we define the firm's markup as $\mu = \frac{\varepsilon}{\varepsilon-1}$. Linearizing around the steady-state $\pi_t = \bar{\Pi}$,

$$\ln(1 + \pi_t) = \frac{\varepsilon}{\theta_p} \left(mc_t - \frac{1}{\mu} \right) + \frac{1}{1 + r_{t+1}} \ln(1 + \pi_{t+1}) \frac{Y_{t+1}}{Y_t},$$

where $\kappa = \varepsilon/\theta_p$ and $mc_t = w_t/Z_t$. This environment is equivalent to Calvo (1983) in first order around zero inflation.

Wage Phillips Curve

We will use the same setting as Hagedorn, Manovskii, and Mitman (2019), but considering just the formal sector. Assume there is a union that aggregate effective labor input into

$$h_t = \left(\int_0^1 e_{i,t}(h_{i,t})^{\frac{\varepsilon_w-1}{\varepsilon_w}} di \right)^{\frac{\varepsilon_w}{\varepsilon_w-1}},$$

sells households labor services $e_{i,t}h_{i,t}$ at nominal wage $W_{i,t}$ to the labor recruiter, which minimizes costs $\int_0^1 W_{i,t}e_{i,t}h_{i,t} di$, implying a demand of

$$h_{i,t} = \left(\frac{W_{i,t}}{W_t} \right)^{-\varepsilon_w} h_t.$$

The unions sets a nominal wage $\hat{W}_t$ for an effect unit of labor, $W_{i,t} = \hat{W}_t$ to maximize profits subject to Rotemberg (1982)'s wage adjustment costs

$$e_{i,t} \frac{\theta_w}{2} \left(\frac{\hat{W}_t}{\hat{W}_{t-1}} - \bar{\Pi}^w \right)^2 h_t.$$

We assume that the union's wage setting problem is

$$V_t^w = \max_{\hat{W}_t} \int \left[\frac{(1 - \tau_\ell)\hat{W}_t}{P_t} e_{i,t}h_{i,t} - \frac{v(h_{i,t})}{u'(\tilde{C}_t)} - e_{i,t}h_t \frac{\theta_w}{2} \left(\frac{\hat{W}_t}{\hat{W}_{t-1}} - \bar{\Pi}^w \right)^2 \right] d\Lambda + \bar{\beta}V_{t+1}^w.$$

Using the same algebra as the price Phillips curve, assuming that the union allocates the same amount of hours in each worker $h_{i,t} = h_t$, and using $\int_F e_{i,t}d\Lambda = L_t/h_t$ and $\int_F d\Lambda = F_t$, it is possible to find that

$$\theta_w(\pi_t^w - \bar{\Pi}^w)\pi_t^w = \varepsilon_w \left[\frac{F_t h_t}{L_t} \frac{v'(h_t)}{u'(\tilde{C}_t)} - \frac{\varepsilon_w - 1}{\varepsilon_w} (1 - \tau_\ell) w_t \right] + \bar{\beta} \theta_w (\pi_{t+1}^w - \bar{\Pi}^w) \pi_{t+1}^w \frac{h_{t+1}}{h_t},$$

where $L_t/(h_t F_t) = \mathbb{E}(e_{i,t}|F)$. We can linearize around the steady state $\pi_t^w = \bar{\Pi}^w$ and take $h_{t+1} = h_t$, since there is no population nor technological growth. Thus,

$$\ln(1 + \pi_t^w) = \frac{\varepsilon_w}{\theta_w} \frac{F_t h_t}{L_t} \left[\frac{v'(h_t)}{u'(\tilde{C}_t)} - (1 - \tau_\ell) \frac{w_t L_t}{h_t F_t} \frac{\varepsilon_w - 1}{\varepsilon_w} \right] + \bar{\beta} \ln(1 + \pi_{t+1}^w).$$

Therefore, defining $\kappa_w = (\varepsilon_w/\theta_w) \cdot Fh/L$ at the steady state and $\mu_w = \varepsilon_w/(\varepsilon_w - 1)$,

the wage Phillips curve is

$$\ln(1 + \pi_t^w) = \kappa_w \left[\psi \cdot h_t^{1/\varphi} \cdot \tilde{C}_t^{1/\gamma} - \frac{(1 - \tau_\ell)w_t L_t}{\mu_w h_t F_t} \right] + \bar{\beta} \ln(1 + \pi_{t+1}^w).$$